RAJALAKSHMI ENGINEERING COLLEGE

Janarthan R 250701261-2025-CSE

J2

Started on	Tuesday, 25 August 2026, 12:10 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:18 PM
Time taken	7 mins 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include<stdio.h>

void swap(int *a,int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[],int low,int high){
    int i = low - 1;
    int pivot = arr[high];
    for(int j = low;j <= high;++j){
        if(arr[j] < pivot){
            ++i;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[i + 1],&arr[high]);
    return i+1;
}

void quicksort(int arr[],int low,int high){
    if(low < high){
        int p = partition(arr,low,high);
        quicksort(arr,low,p - 1);
        quicksort(arr,p+1,high);
    }
}

int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;++i) scanf("%d",&arr[i]);
    int low = 0,high = n - 1;
    quicksort(arr,low,high);
    for(int i=0;i<n;++i){
        printf("%d ",arr[i]);
    }
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔

	Input	Expected	Got	
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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